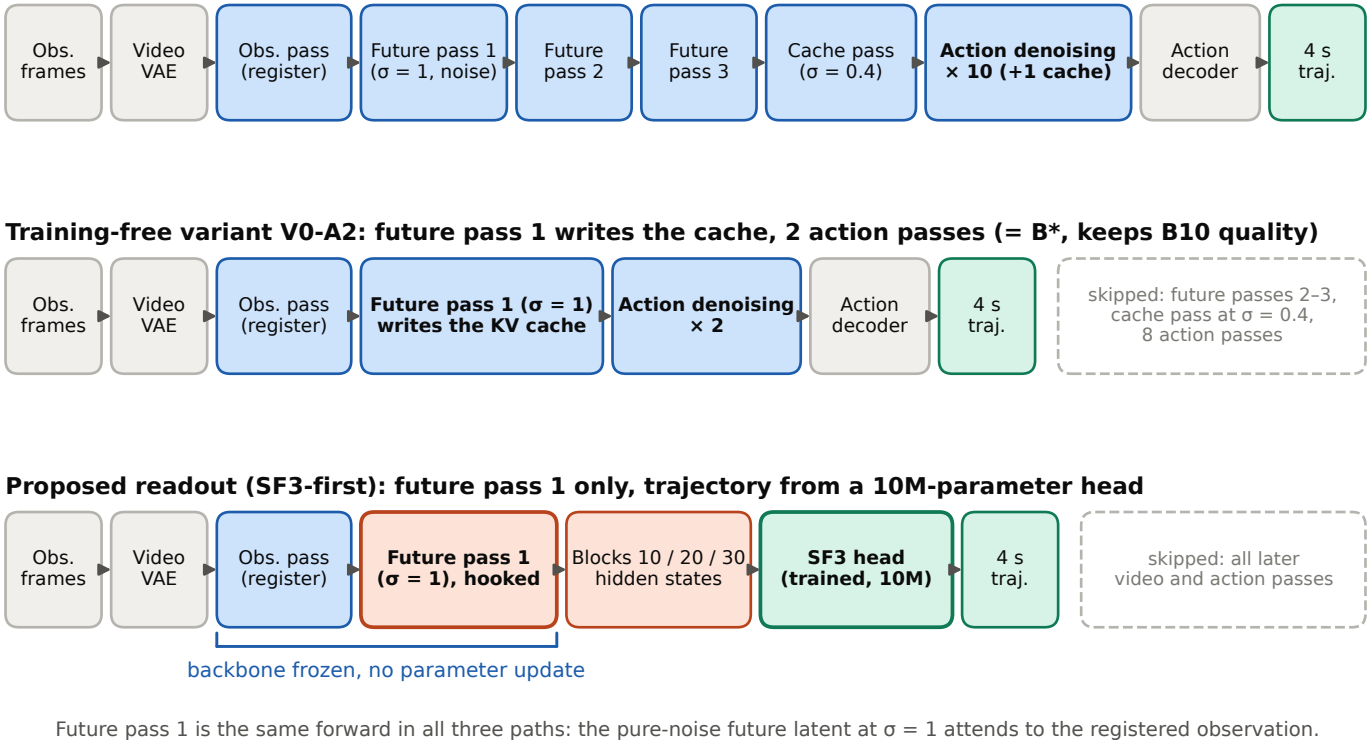
DriveWAM, official inference (B10): 4 future passes, 10 action passes
Obs. frames
Video VAE
Obs. pass (register)
Future pass 1 (σ = 1, noise)
Future pass 2
Future pass 3
Cache pass (σ = 0.4)
Action denoising × 10 (+1 cache)
Action decoder
4 s traj.
Training-free variant V0-A2: future pass 1 writes the cache, 2 action passes (= B*, keeps B10 quality)
Obs. frames
Video VAE
Obs. pass (register)
Future pass 1 (σ = 1) writes the KV cache
Action denoising × 2
Action decoder
4 s traj.
skipped: future passes 2–3, cache pass at σ = 0.4, 8 action passes
Proposed readout (SF3-first): future pass 1 only, trajectory from a 10M-parameter head
Obs. frames
Video VAE
Obs. pass (register)
Future pass 1 (σ = 1), hooked
Blocks 10 / 20 / 30 hidden states
SF3 head (trained, 10M)
4 s traj.
skipped: all later video and action passes
backbone frozen, no parameter update
Future pass 1 is the same forward in all three paths: the pure-noise future latent at σ = 1 attends to the registered observation.